



Automation & Development of Calibration & Maintenance Computerized Management Information System (CMCMIS)

TIMCD-EFMG-ESS
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MAJOR OBJECTIVES:

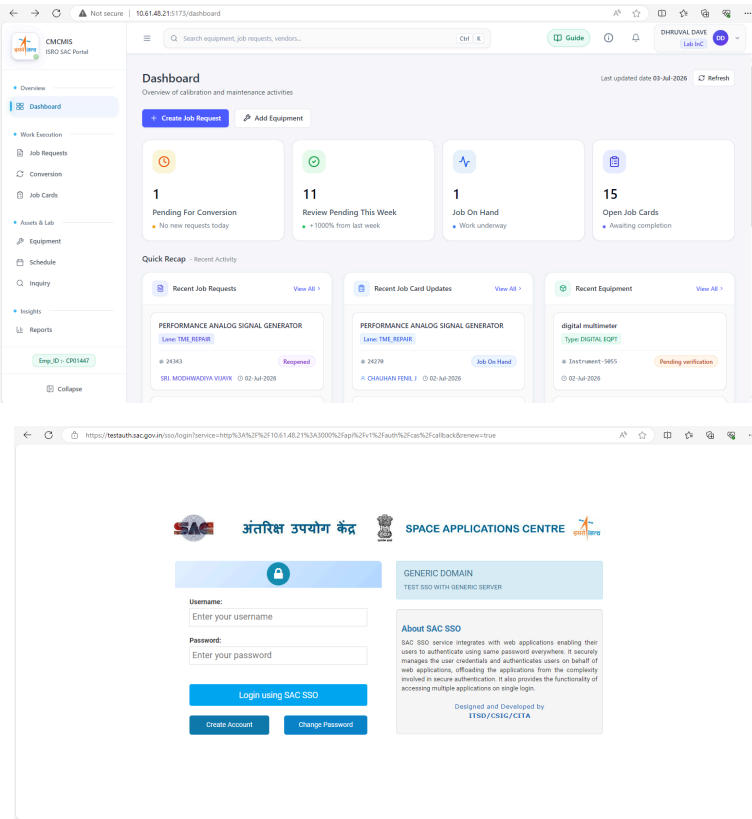
- Digitize the end-to-end Job Request → Job Card → Certificate lifecycle within a single role-aware web application.
- Implement Role Based Access Control (Super Admin, Lab In-Charge, Lab Engineer, Normal User, View-Only) with lane-level data scoping.
- Automate generation of NABL-accredited and Non-NABL calibration certificates, Job Closing Forms, and JRFs as PDF.
- Preserve and extend the pre-existing legacy MySQL master schema without discarding historical data.
- Provide real-time dashboards, operational reports, procurement management, scheduling, and a complete audit trail.

METHODOLOGY FLOW CHART:

Requirements elicitation & domain study
→ UI/UX wireframing & high-fidelity design
→ Project scaffolding, DB migrations & authentication
→ Job Request & Job Card state-machine workflows
→ Checklists, spares, documents & PDF certificate generation
→ Procurement, lab scheduling & in-app notifications
→ Reports, analytics, dashboard Key Performance Indicators & audit trail
→ Integration testing, bug fixes & documentation

RESULTS/MAJOR FINDINGS:

- Delivered a full-stack web app: React 18 + Vite frontend, Node.js/Express backend (23 modules), MySQL with 40+ table schema.
- Complete Job Request → Job Card lifecycle with Calibration, Maintenance & Repair sub-workflows, task checklists, and spares tracking.
- Multi-factor secure login and SSO Implementation for employee authentication.
- NABL & Non-NABL certificates, Job Closing Forms and JRFs generated as PDFs in seconds.
- Five-role RBAC (Super Admin, Lab In-Charge, Lab Engineer, Normal User, View-Only) enforced on every endpoint with lane-level data scoping.
- Six canned operational reports (Calibration-Due, Pending Jobs, Equipment Utilization, Engineer Summary, Job-Card & Job-Request Summary) – on-screen and PDF export.
- Real-time equipment calibration-due register with bulk-done marking, replacing a manually updated spreadsheet.
- Full field-level audit trail logging every action across all modules – actor, timestamp, and before/after values captured.
- Lab scheduling with ICS export, procurement (POs + spare-parts inventory), in-app notifications and cross-entity quick-search delivered as part of the complete operational suite.



CONCLUSION: The internship at SAC-ISRO provided end-to-end ownership of a real-world software project – from requirements and design through full-stack implementation, testing, and documentation. The delivered CMCMIS system completely digitises the calibration and maintenance workflow at TIMCD, introducing role-based access control, automated NABL-compliant PDF generation, Secured multi-factor authentication with SAC-SSO, and a full audit trail. The experience sharpened practical skills in full-stack development, database design, and security engineering directly applicable to production software in a government-research environment.